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Date: 1/15/24

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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Find the slope of the function's graph at the given point. Then find an equation for the line tangent to the graph there.

$$h(t) = t^3, (5, 125)$$

The slope of the curve $y = f(t)$ at the point $P(t_0, f(t_0))$ is determined by the equation below, provided the limit exists.

$$m = \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h}$$

The tangent line to the curve at P is the line through P with this slope.

Evaluate the equation for the curve for t_0 and $t_0 + h$ using $t_0 = 5$, and substitute the results into the expression in the limit. Multiply out the numerator. Use the fact that $(t + h)^3 = t^3 + 3ht^2 + 3h^2t + h^3$.

$$m = \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(5 + h)^3 - (5)^3}{h}$$

Substitute.

$$= \lim_{h \rightarrow 0} \frac{5^3 + 3h \cdot 5^2 + 3h^2 \cdot 5 + h^3 - (5)^3}{h}$$

Cube the expression.

Now simplify the numerator.

$$m = \lim_{h \rightarrow 0} \frac{5^3 + 3h \cdot 5^2 + 3h^2 \cdot 5 + h^3 - (5)^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{75h + 15h^2 + h^3}{h}$$

Simplify.

Since h approaches 0 but $h \neq 0$, cancel h from the numerator and denominator. Evaluate the limit by substituting $h = 0$.

$$\lim_{h \rightarrow 0} \frac{75h + 15h^2 + h^3}{h} = \lim_{h \rightarrow 0} \frac{h(75 + 15h + h^2)}{h}$$

Factor the numerator.

$$= \lim_{h \rightarrow 0} (75 + 15h + h^2)$$

Cancel the h 's.

$$= 75 + 15(0) + (0)^2$$

Substitute.

$$= 75$$

Simplify.

The slope m of the graph of the function $h(t) = t^3$ at $(5, 125)$ is 75.

Since the slope of the tangent line is the same and a point on the line is given, use the point-slope equation $y = y_0 + m(t - t_0)$ to determine the equation of the tangent line.

$$y = y_0 + m(t - t_0)$$

$$= 125 + (75)(t - 5)$$

Substitute.

$$= 75t - 250$$

Distribute and simplify.

The line $y = 75t - 250$ is tangent to the graph of the function $h(t) = t^3$ at the point $(5, 125)$.